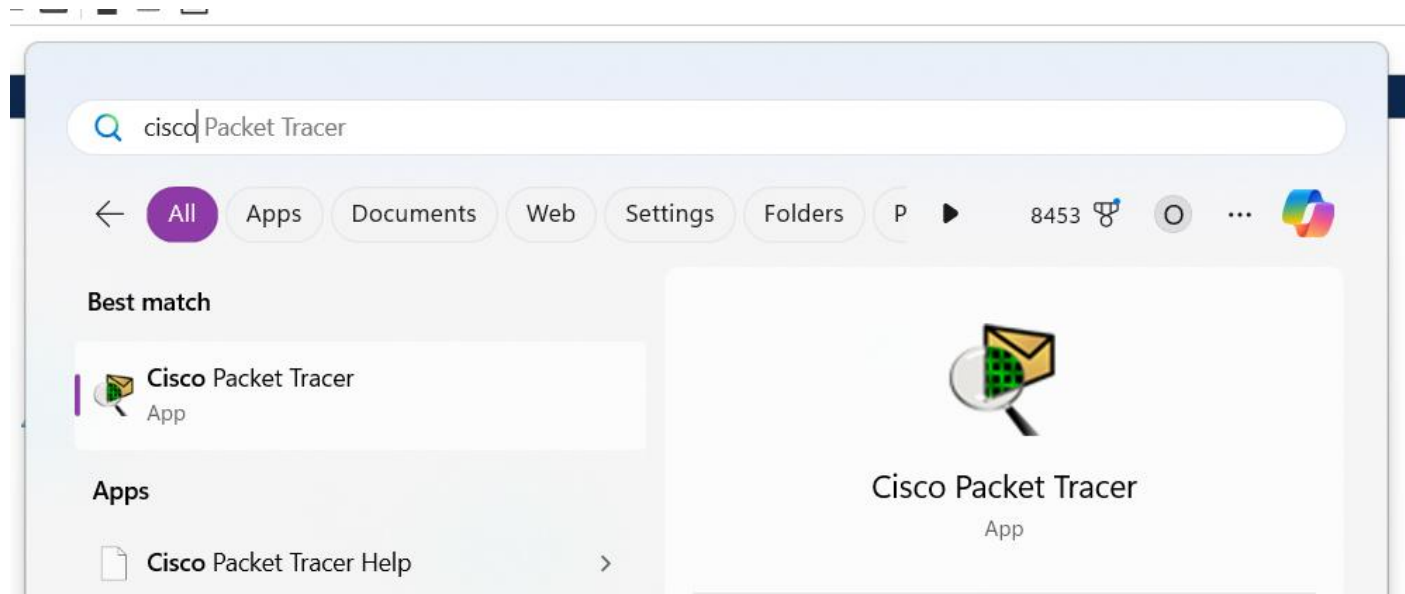
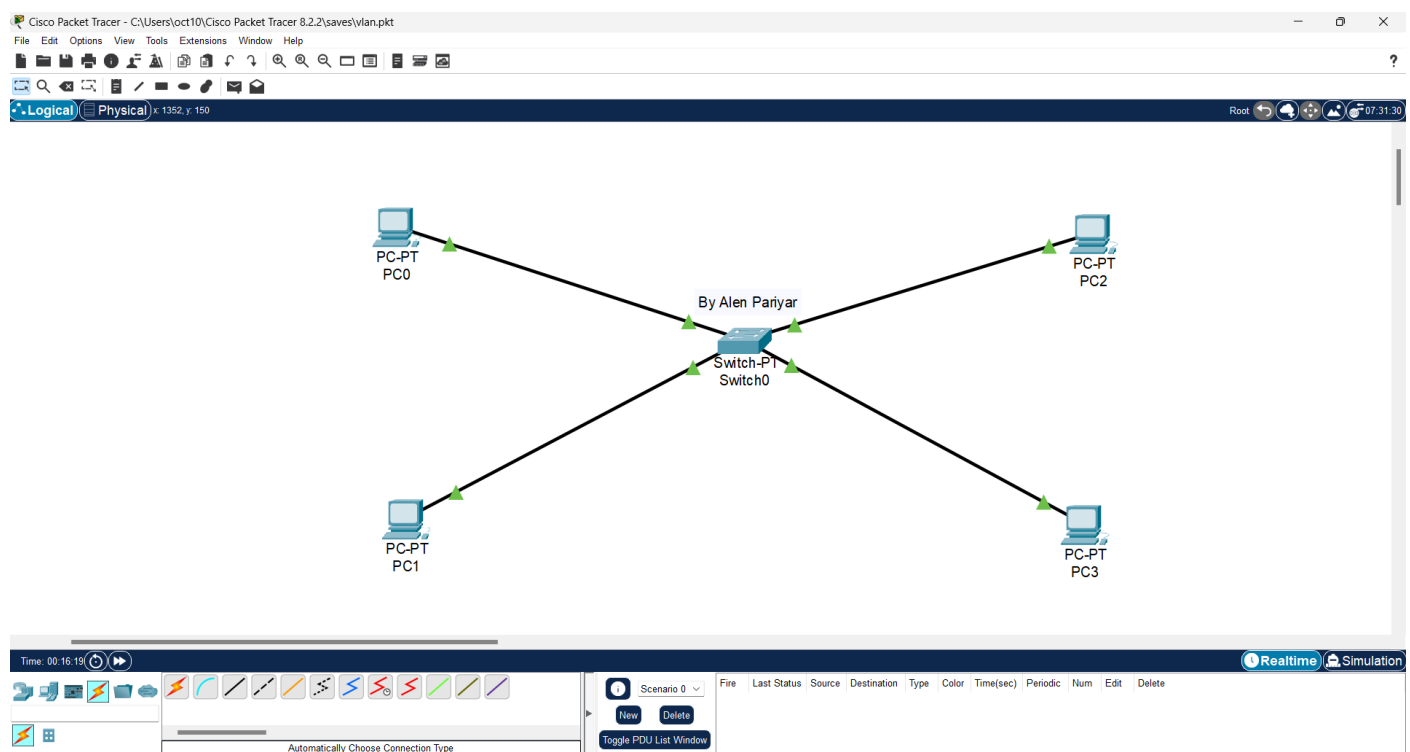


## Demonstrate the use of VLAN

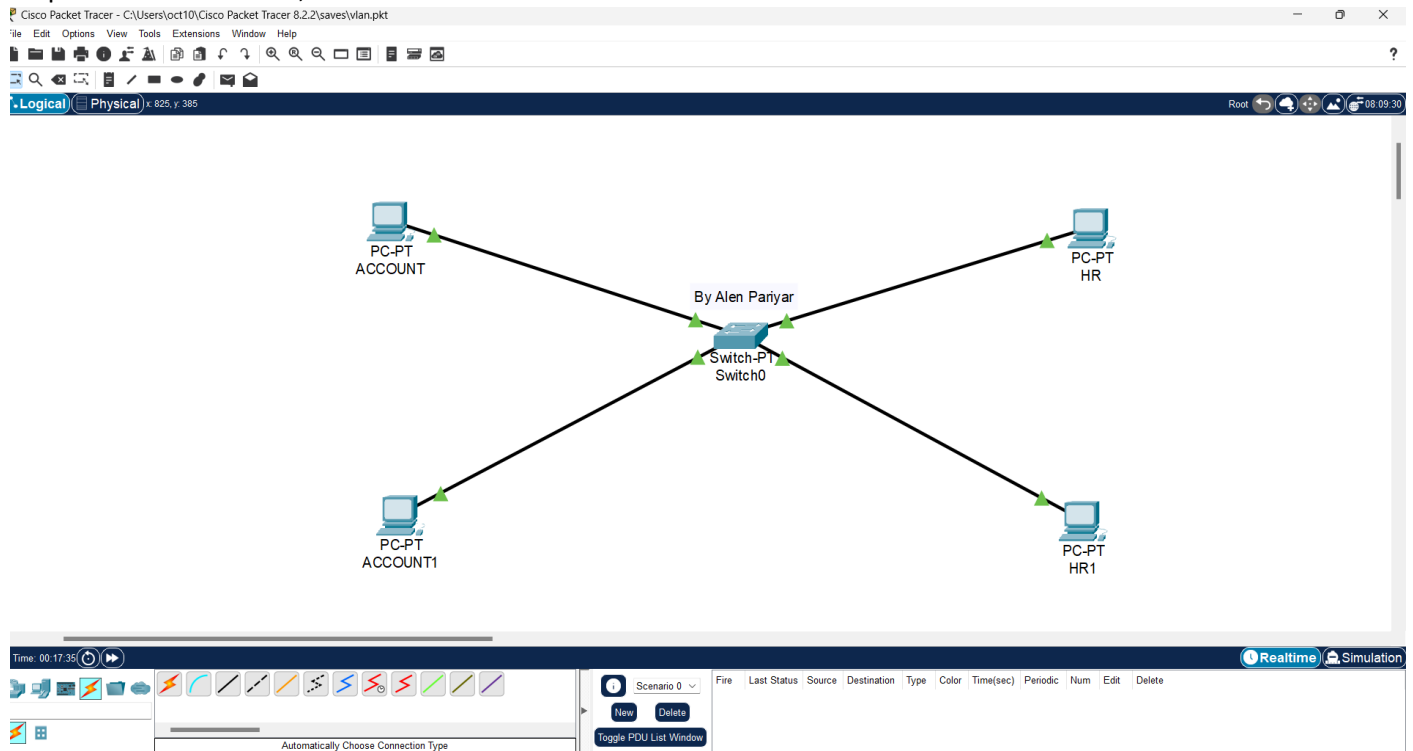
Step 1 : Open Cisco Packet Tracer,



Step 2 : Drag and Drop : PC's and Switch and connect them using suitable wire,

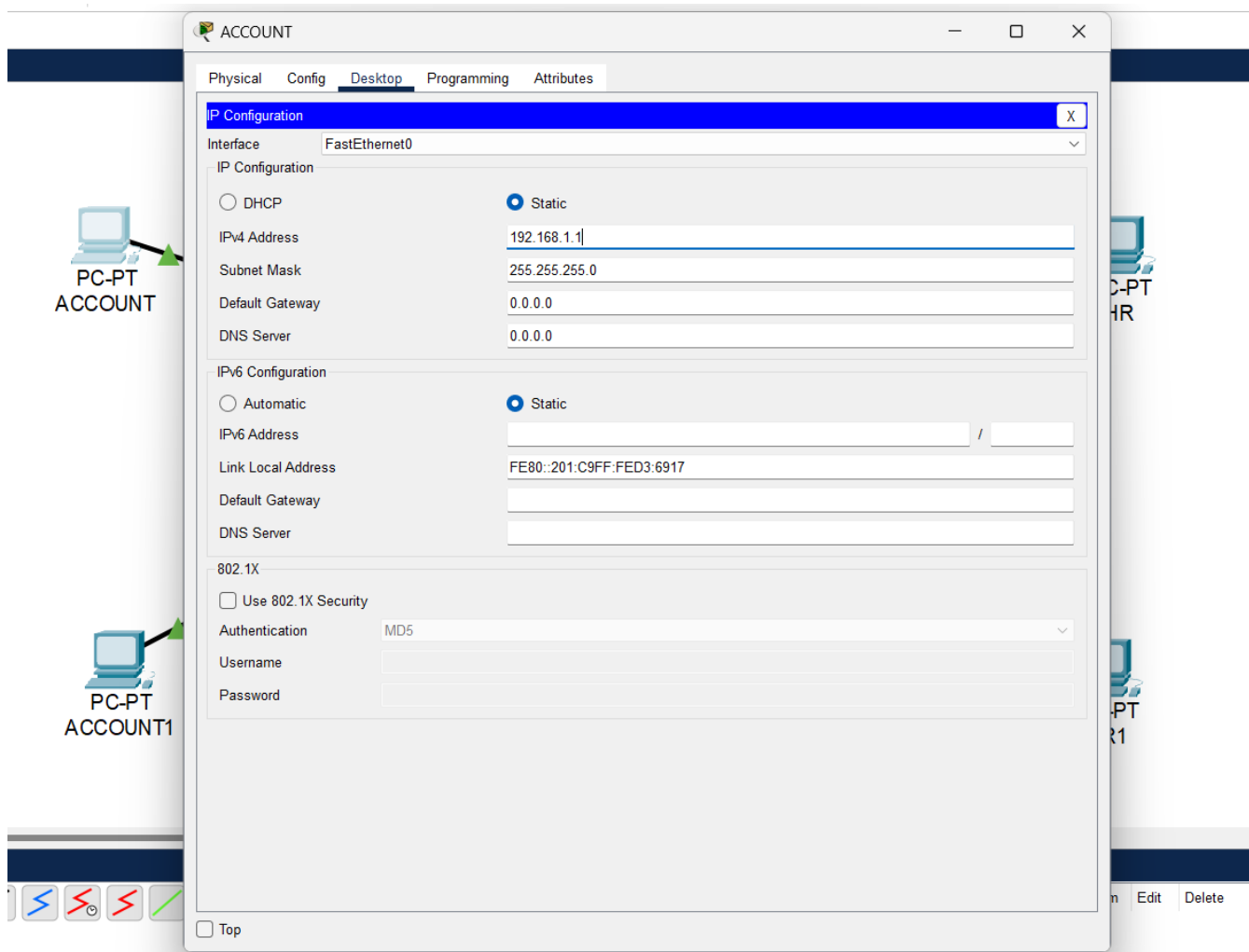


### Step 3: Rename the PC's,



### Step 4: Setup and Configure IP address of each PC,

- Click on PC icon,
- Click on Desktop Tab
- Click on "IP Configuration"



### Step 5: Setup VLAN Configuration,

- Click on Switch icon,
- Click on Config Tab,
- Click on **VLAN Database**,
- Enter VLAN Number and VLAN Name and Click on Add,

Switch0

Physical **Config** CLI Attributes

**GLOBAL**

Settings

Algorithm Settings

**SWITCHING**

**VLAN Database**

**INTERFACE**

FastEthernet0/1

FastEthernet1/1

FastEthernet2/1

FastEthernet3/1

FastEthernet4/1

FastEthernet5/1

VLAN Configuration

VLAN Number: 20

VLAN Name: HR

Add Remove

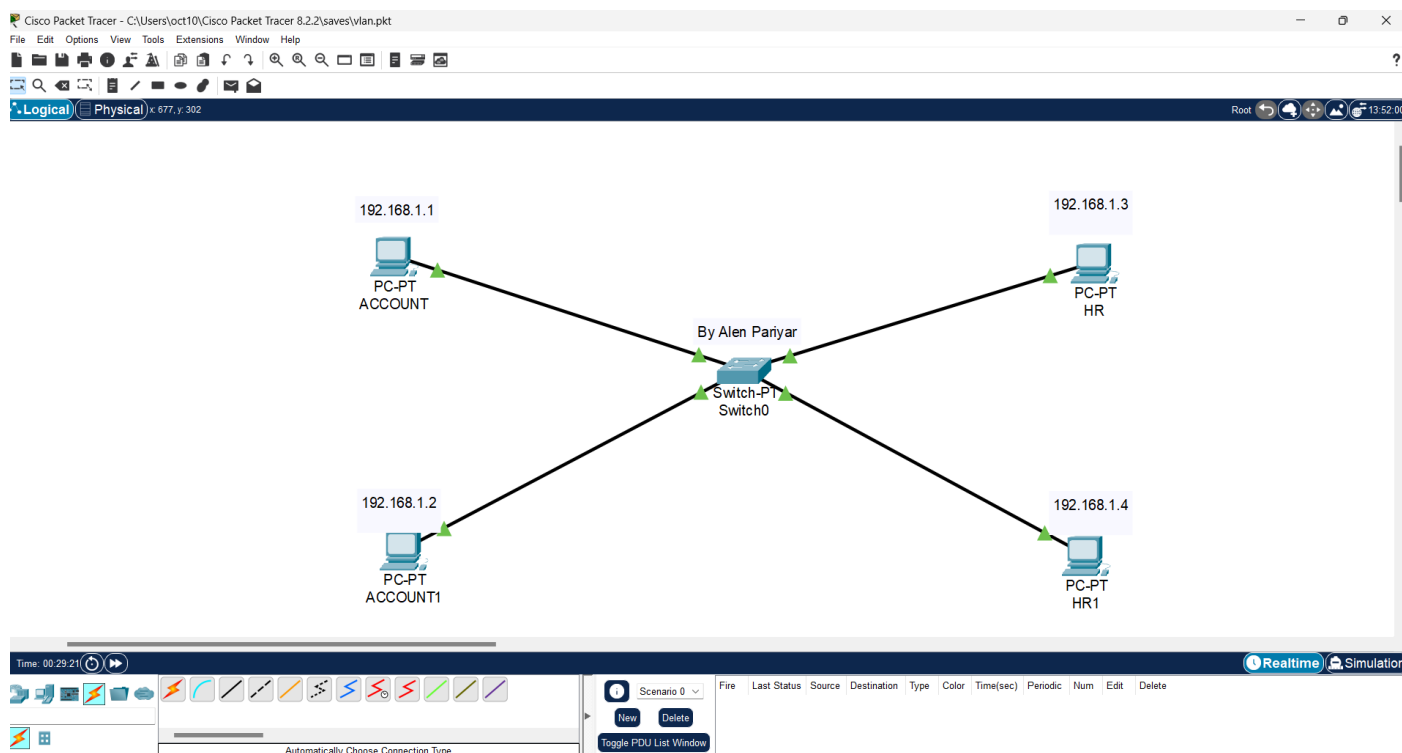
| VLAN No | VLAN Name          |
|---------|--------------------|
| 1       | default            |
| 10      | Account            |
| 20      | HR                 |
| 1002    | fddi-default       |
| 1003    | token-ring-default |
| 1004    | fddinet-default    |
| 1005    | trnet-default      |

Equivalent IOS Commands

```
Switch(config-if)#exit
Switch(config)#interface FastEthernet2/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name Account
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)#
```

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Step 6 : Check PING command in PC's between different departments,



(PING from ACCOUNT to HR1) is getting reply,

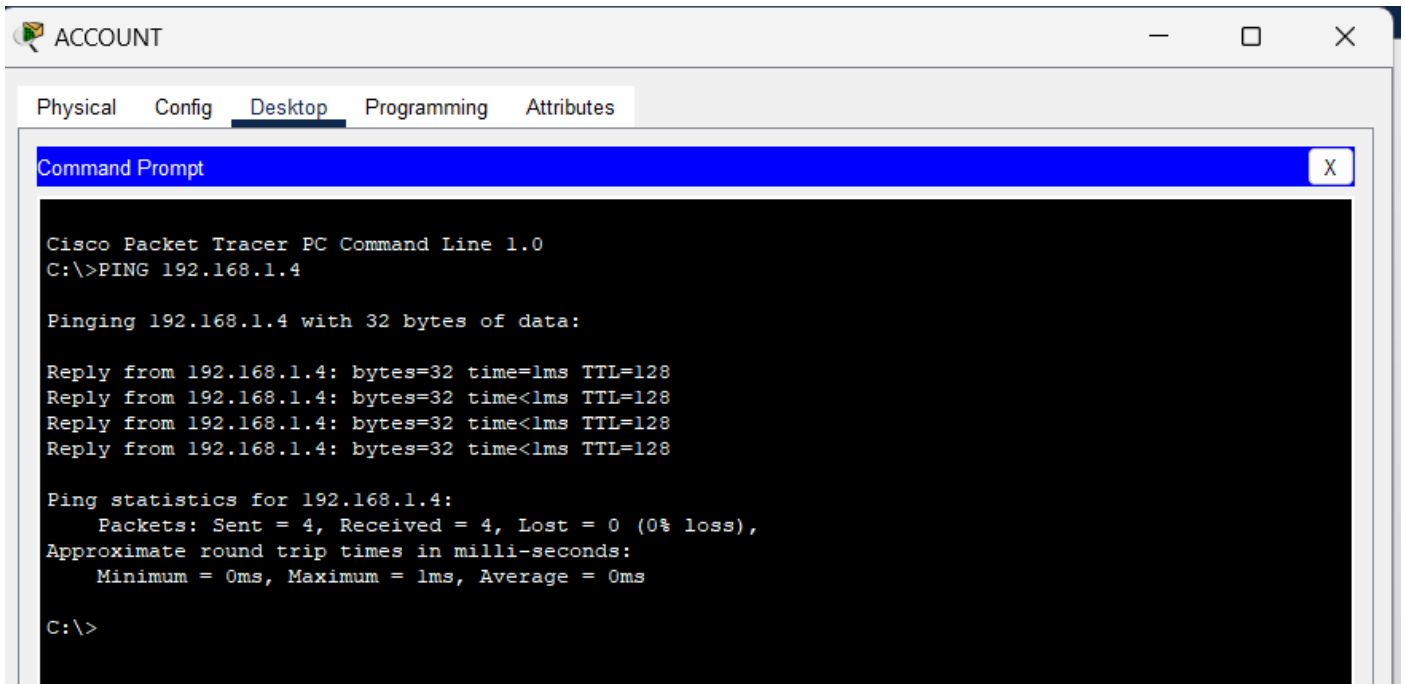
```
Cisco Packet Tracer PC Command Line 1.0
C:\>PING 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

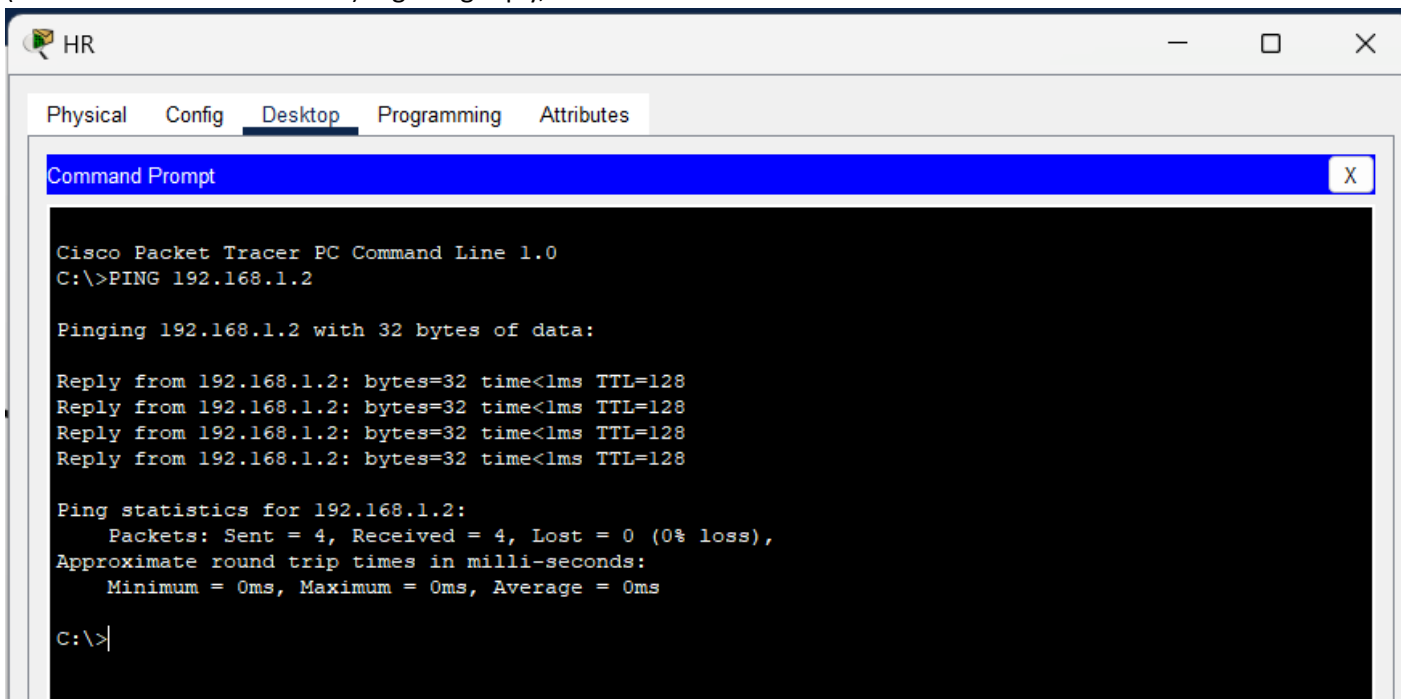
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

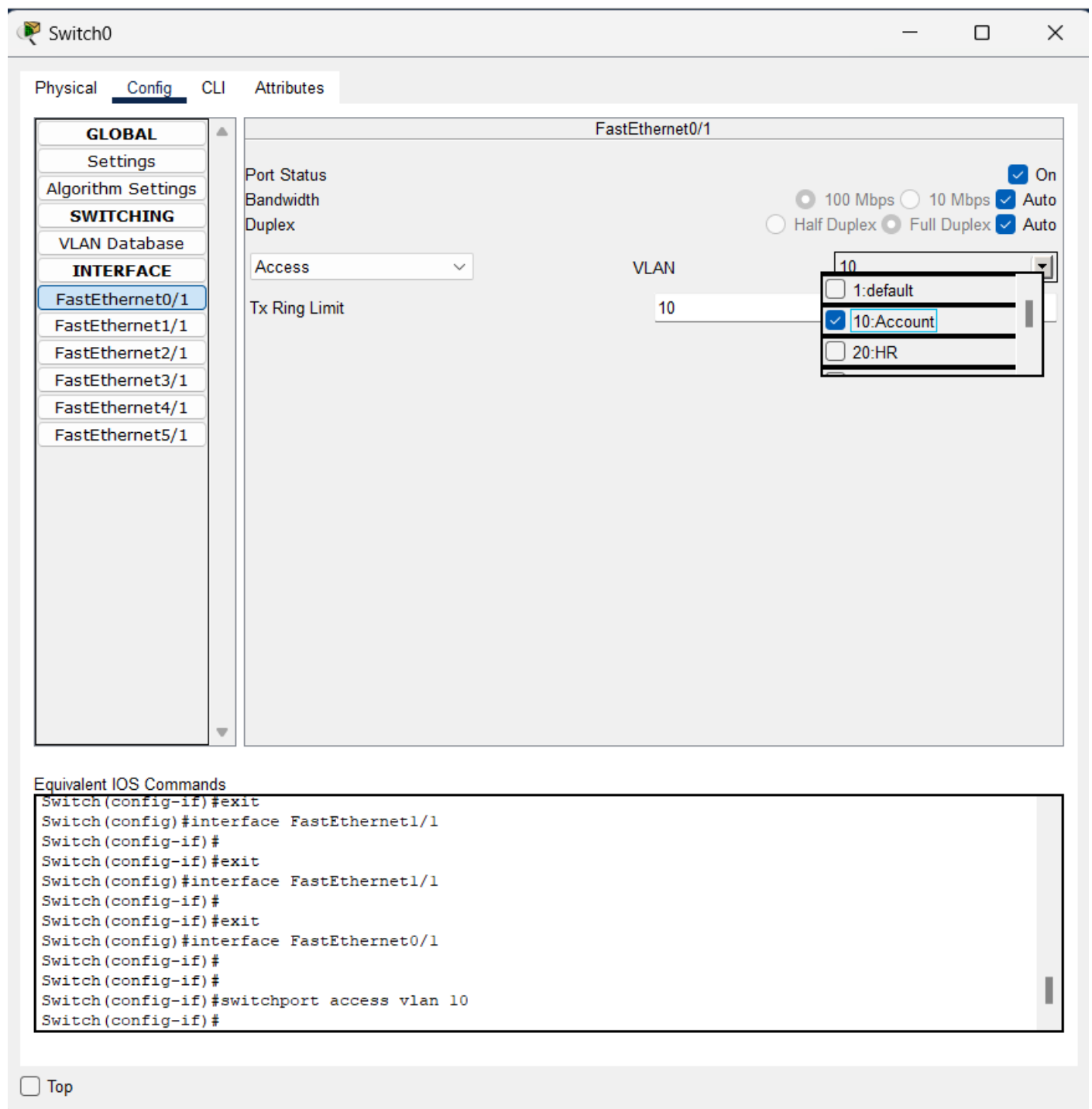
C:\>|
```



(PING from HR to ACCOUNT1) is getting reply,



Step 7: Assign VLAN to each **INTERFACE** of Switch connected to each **PC**,



Switch0

Physical **Config** CLI Attributes

**GLOBAL**

- Settings
- Algorithm Settings

**SWITCHING**

- VLAN Database

**INTERFACE**

- FastEthernet0/1**
- FastEthernet1/1
- FastEthernet2/1
- FastEthernet3/1
- FastEthernet4/1
- FastEthernet5/1

FastEthernet0/1

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

Access  VLAN

☐ 1:default

☒ 10:Account

☐ 20:HR

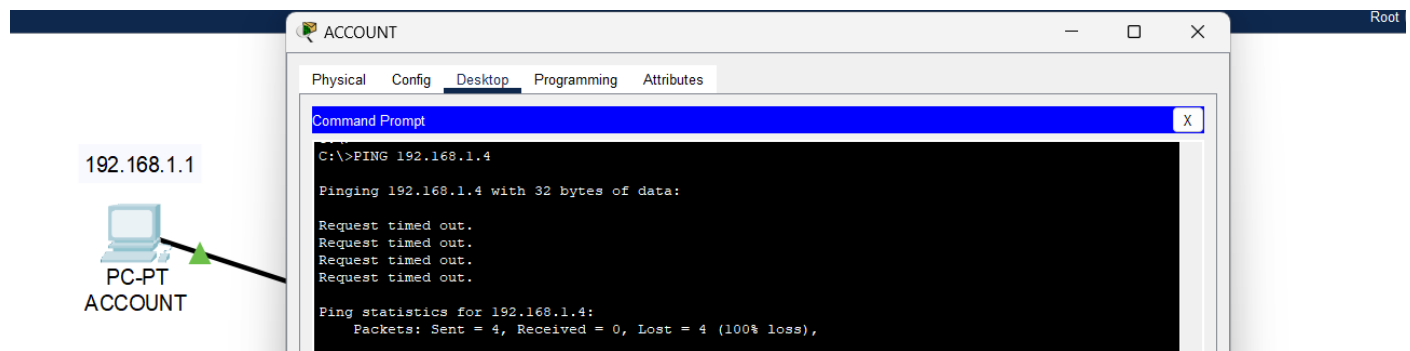
Tx Ring Limit

Equivalent IOS Commands

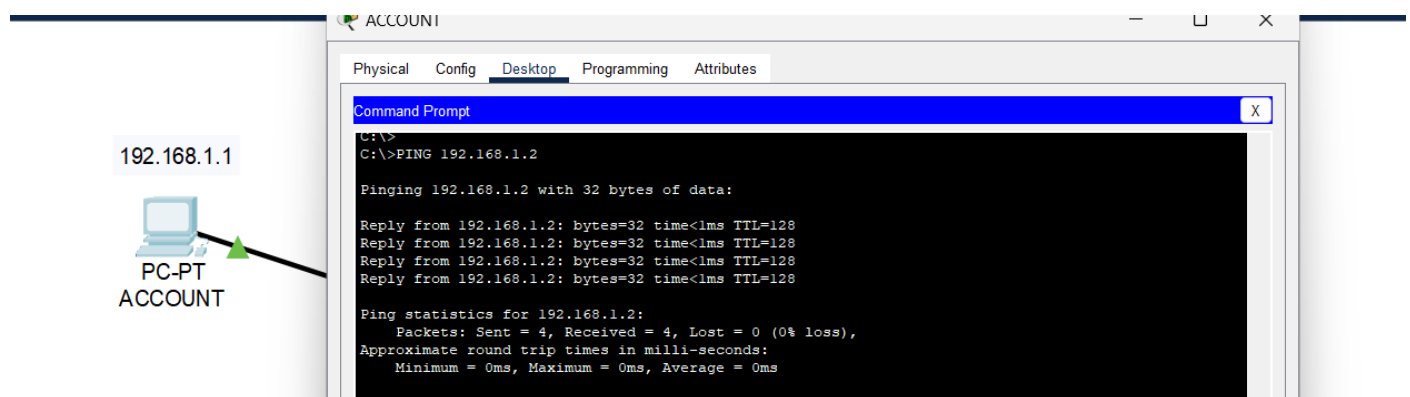
```
Switch(config-if)#exit
Switch(config)#interface FastEthernet1/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet1/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
```

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Step 7 : Check PING command in PC's between departments after assigning VLAN,



(PING from ACCOUNT to HR1) is not getting Reply.



(PING from ACCOUNT to ACCOUNT1) is getting Reply,